## Answer Key & Explanations

1. 2.0 Amps
    a. $R_{parallel} = (6 \times 3) / (6 + 3) = 18/9 = 2 \Omega$.
    b. $R_{total} = 4 \text{ (series)} + 2 \text{ (parallel part)} = 6 \Omega$.
    c. $I = V/R = 12V / 6 \Omega = 2A$.
2. 4.0 Volts
    a. The total current (2A) flows through the parallel bank.
    b. $V_{parallel} = I_{total} \times R_{parallel} = 2A \times 2 \Omega = 4V$. (Since B is in that parallel bank, it sees 4V).
3. 5.33 Watts
    a. $P = V^2 / R$. Voltage across C is 4V (same as B).
    b. $P = 4^2 / 3 = 16 / 3 = 5.33 W$.
4. 0.59 $\Omega$
    a. $V_{terminal} = EMF - I(r)$.
    b. Current $I = V_{load} / R_{load} = 8.5V / 10 \Omega = 0.85A$.
    c. $8.5V = 9.0V - (0.85A \times r)$.
    d. $0.5V = 0.85r \rightarrow r \approx 0.588 \Omega$.
5. 4800 $\mu C$ (or 4.8 mC)
    a. Parallel Capacitors add: $C_{total} = 100 + 300 = 400 \mu F$.
    b. $Q = C \times V = 400 \mu F \times 12V = 4800 \mu C$.
6. 12 Volts
    a. Voltage Divider Rule: $V_{R2} = V_{total} \times [R_2 / (R_1 + R_2)]$.
    b. $15V \times [4k / (1k + 4k)] = 15 \times (4/5) = 15 \times 0.8 = 12V$.
7. $0.36
    a. Energy = Power $\times$ Time = $0.1kW \times 24h = 2.4 kWh$.
    b. Cost = $2.4 kWh \times \$0.15 = \$0.36$.
8. False (Conventional flows + to -, electrons flow - to +).
9. Decreases by factor of 9 ($1/r^2$ law).
10. South Pole (Right Hand Grip Rule: Curl fingers clockwise, thumb points away from you (into the coil). The face you see is South).
11. 12 Volts
    a. Ratio $500:50$ is $10:1$ Step Down.
    b. $120V / 10 = 12V$.
12. Zero (To not impede current flow).
13. Parallel (To measure potential difference across the component).
14. Counter-Clockwise (Thumb points up, fingers curl CCW).

15. Thomas Edison
16. GFCI (Ground Fault Circuit Interrupter)
17. Higher
18. Anode (Oxidation is Loss of electrons).
19. Lose Electrons
20. South
    a. Rule: $F = qv \times B$.
    b. Velocity: East. B-Field: Down.
    c. $v \times B$ (East crossed Down) = North.
    d. However, electron is negative. Flip the direction.
    e. Force is South.
21. Variable Resistor (Potentiometer/Rheostat)
22. They all go out (Open circuit).
23. AND Gate
24. Shorted
25. SPDT (Single Pole Double Throw) - also called "Three-Way Switches" in household wiring.
26. Connected
27. Lower (The voltmeter acts as a parallel resistor, lowering equivalent resistance and thus the voltage drop at that point).
28. The longer leg is the anode (positive +), and the shorter leg is the cathode (negative -)
29. No
    a. $I = P / V = 600W / 120V = 5A$.
    b. It is exactly at the rating. Usually, fuses blow *above* the rating. It will run, but it's on the edge. (Technically "No, but it's close").
30. False (You get two smaller magnets, each with N and S).
31. C
32. B
33. D
34. B
35. D